

Robot Navigation Anomaly Detection

Team 2 | AI Algorithms | Mid-term Presentation | January 6, 2026

Agenda

- 1. Requirements & Changes (~1 min)
- 2. Metrics Overview (~2 min)
- 3. Interesting Insights from Anomaly Detection (~3 min)
- 4. Methodology for Causal Analysis & Results (~3 min)
- 5. Ideas for Future Improvements (~1 min)

1. Requirements & Changes

~1 min

Initial Approach (Problem)

- Log-based anomaly detection
- Dataset lacks RTF (Real-Time Factor) data
- Logs were not structured for analysis

Current Approach (Solution)

- CSV-based anomaly detection
- Structured, reliable data format
- Consistent across all runs

Data Sources

~1 min

- poses.csv — Ground truth and estimated poses
- behaviors.csv — Behavior tree execution logs
- rosbag2.csv — Raw ROS topics (AMCL pose, transforms, velocity commands)
- scenario.config — Scenario parameters

2. Performance Metrics

~2 min

Category	Metric	Unit	Description
Localization	mean_pos_error	m	Average position error
Localization	rmse_pos	m	Root mean square error (spike sensitive)
Localization	mean_yaw_error	rad	Average orientation error
Path	executed_path_length	m	Total path traveled
Path	path_efficiency	ratio	gt_path / executed (1 is best)
Kinematics	mean_linear_velocity	m/s	Average movement speed
Kinematics	trajectory_smoothness	rad/s ²	Angular acceleration (lower = smoother)
Kinematics	duration	s	Total run time
AMCL	mean_amcl_uncertainty	m	Average localization confidence

Rule-Based (8 types)

- goal_failure - Goal not reached
- no_initiation - Robot didn't start
- position_error_spike - Sudden errors
- stuck - Robot stopped moving
- high_amcl_uncertainty - Poor localization
- high_yaw_error - Orientation issues
- path_inefficiency - Sub-optimal path
- oscillation - Jerky motion (smoothness > 2.0)

ML-Based (1 type)

- Isolation Forest
- Captures multi-metric abnormal patterns that rules may miss.
- Uses all performance metrics

3. Interesting Insights from Anomaly Detection

~3 min

Anomaly Type	Count	% of Runs
goal_failure	150	50.0%
stuck	115	38.3%
Isolation Forest	45	15.0%
position_error_spike	15	5.0%
high_yaw_error	9	3.0%
path_inefficiency	7	2.3%

Key Findings

- Total occurrences: 341
- goal_failure is most common (50%)
- stuck follows at 38.3%
- Isolation Forest catches 15%
- Single run can have multiple anomalies

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- Door-Width (120 Runs): High incidence of Goal Failure (64% of runs).
 - Hallway-Window (120 Runs): Goal Failure remains the top anomaly (43% of runs).
 - Room-Size (60 Runs): Agents frequently getting Stuck (52% of runs).

Insights - Anomaly Co-occurrence

~3 min

- position_error_spike + high_yaw_error (Jaccard = 0.60)
 - -> Poor orientation tracking leads to position drift
- high_yaw_error + path_inefficiency (Jaccard = 0.78)
 - -> Orientation errors cause inefficient path execution

Insights - Early Detection (First 30% of Run)

~3 min

Anomaly	PR-AUC	ROC-AUC	Cases	Insight
goal_failure	0.852	0.828	150	Highly predictable early
stuck	0.766	0.789	115	Highly predictable early
Isolation Forest	0.661	0.834	45	Good detection capability

Insights - Feature Signatures

~3 min

Goal Failure :

path_length	: $r=-0.491$ ***
std_velocity	: $r=-0.437$ ***
path_efficiency	: $r=-0.384$ ***

Stuck :

duration	: $r=+0.524$ ***
mean_angular_vel	: $r=-0.283$ ***
path_length	: $r=+0.276$ ***

high_amcl_uncertainty:

rmse_pos : r=+0.291 ***

max_velocity : r=+0.277 ***

std_pos_error : r=+0.274 ***

4. Causal Analysis - (Ensemble + Model Distillation / Surrogate Modeling)

- Step 1: High-Capacity Model Training (Ensemble)
 - -> Voting Classifier: Decision Tree+ Random Forest + Gradient Boosting
 - -> Prioritizes F1-score
- Step 2: Surrogate Model Training (Distilled Decision Tree)
 - -> Constrained tree (max_depth=4)
 - -> Target: Ensemble predictions (not ground truth)

Step 3: Rule Extraction

- -> Extract paths with $P(\text{Anomaly}) > 0.5$
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- Step 4: FOL Translation
 - -> Map thresholds to semantic predicates

Derived FOL Rules - Examples

~3 min

Goal Failure:

(Cov=17.0%, Prec=1.00, Rec=0.34, n=51), Confusion: TP=51, FP=0, FN=99, TN=150

FOL: $\forall t : \text{in_corridor}(t) \wedge \text{min_wall_distance}(t) > 0.737 \wedge \text{obstacle_clearance_ratio}(t) \leq 17.970 \Rightarrow \text{goal_failure}$

NL: IF robot is in corridor AND min wall distance > 0.737 AND obstacle clearance ratio $\leq$ 17.970
THEN goal_failure

Stuck:

(Cov=6.7%, Prec=1.00, Rec=0.17, n=20), Confusion: TP=20, FP=0, FN=95, TN=185

FOL: $\forall t : \text{min_door_distance}(t) > 1.792 \wedge \text{noise_level}(t) \leq 0.015 \wedge \text{min_door_distance}(t) > 2.104 \wedge \text{room_area}(t) \leq 25.130 \Rightarrow \text{stuck}$

NL: IF min door distance > 1.792 AND noise level $\leq$ 0.015 AND min door distance > 2.104 AND room area $\leq$ 25.130 THEN stuck

Derived FOL Rules - Examples

High_amcl_uncertainty:

(Cov=4.0%, Prec=1.00, Rec=0.52, n=12), Confusion: TP=12, FP=0, FN=11, TN=277

FOL: $\forall t : \text{obstacle_clearance_ratio}(t) > 31.314 \wedge \text{total_obstacle_area}(t) \leq 0.875 \wedge \text{door_width}(t) > 0.935 \Rightarrow \text{high_amcl_uncertainty}$

NL: IF obstacle clearance ratio > 31.314 AND total obstacle area $\leq$ 0.875 AND door width > 0.935 THEN high_amcl_uncertainty

Path_inefficiency:

(Cov=3.3%, Prec=0.70, Rec=1.00, n=10), Confusion: TP=7, FP=3, FN=0, TN=290

FOL: $\forall t : \text{obstacle_clearance_ratio}(t) \leq 8.833 \wedge \text{clearance_ratio}(t) > 6.812 \wedge \text{min_door_distance}(t) \leq 1.349 \Rightarrow \text{path_inefficiency}$

NL: IF obstacle clearance ratio $\leq$ 8.833 AND clearance ratio > 6.812 AND min door distance $\leq$ 1.349 THEN path_inefficiency

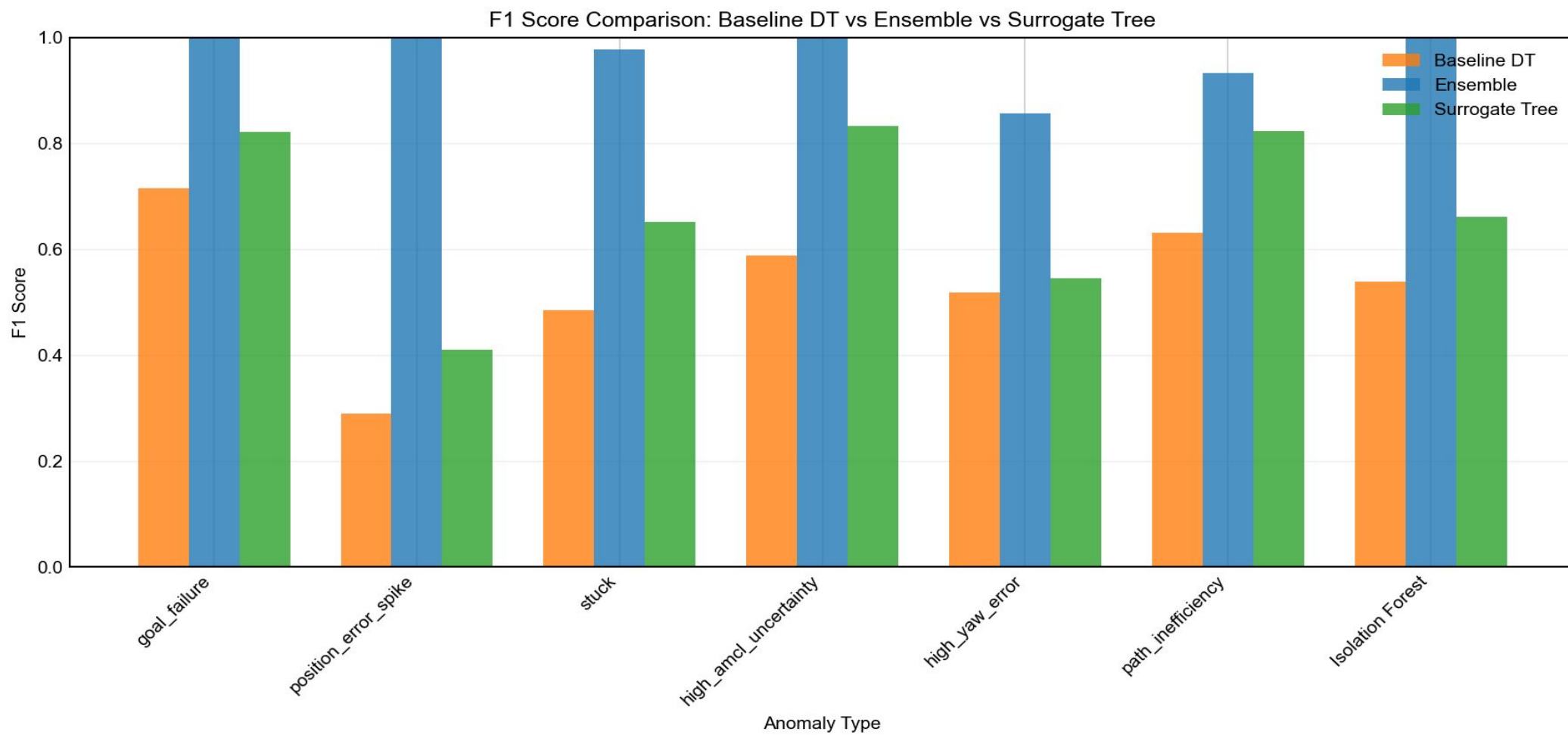
Derived FOL Rules - Examples

Isolation Forest:

(Cov=4.0%, Prec=1.00, Rec=0.27, n=12), Confusion: TP=12, FP=0, FN=33, TN=255

FOL: $\forall t : \text{obstacle_clearance_ratio}(t) > 31.314 \wedge \text{total_obstacle_area}(t) \leq 0.875 \wedge \text{door_width}(t) > 0.935 \Rightarrow \text{Isolation Forest}$

NL: IF obstacle clearance ratio > 31.314 AND total obstacle area $\leq$ 0.875 AND door width > 0.935 THEN
Isolation Forest



5. Ideas for Future Improvements

~1 min

Short-term

- **Hyperparameter tuning:** on surrogate tree
- Optimize max_depth, min_samples_leaf
- Better balance: simplicity vs accuracy

Long-term

- **Feature engineering enhancements:**
 - > Initially we also considered path length but we removed it because of data leakage in the form of reverse causality
 - -> Include temporal dynamics (robot's movement and error metrics change over time. While "Computable Functions" and "Atomic Relations" focus primarily on spatial context) , etc..
- Cross-validation on unseen scenarios
- Sensitivity analysis of rule thresholds

Thank You!